



Shubham Narkhede

PROFESSIONAL SUMMARY

Full-Stack Engineer with 7+ years of experience building real-time visualization platforms and scalable web applications for enterprise systems. Expert in React, Vue.js, Angular, TypeScript, Node.js, and Python (FastAPI), delivering high-performance frontends capable of streaming 10,000+ data points per second at 60 FPS. Experienced in 3D web rendering (Three.js, WebGL), real-time integrations, and low-latency communication using WebSockets and gRPC. Proven track record of owning end-to-end feature delivery from concept to production, automating complex workflows, and mentoring developers in cross-functional teams. Strong in balancing product trade-offs, maintaining high code quality, and improving developer productivity while managing multiple responsibilities simultaneously.

EXPERIENCE

Full Stack Software Engineer Robert Bosch GmbH

07/2022 – Present
Stuttgart-Feuerbach, Germany

- Architected end-to-end full-stack features using TypeScript, React, Angular, Vue.js, Node.js, and Python with scalable modular patterns, establishing frontend standards adopted by 30+ engineers across 5 enterprise platforms.
- Designed and deployed high-volume data visualization dashboards with React, D3.js, and Canvas API, processing 10,000+ data points/second at 60fps for the CPMS, delivering real-time charging session monitoring, live power curves, and station health metric for 30 countries.
- Built real-time geospatial monitoring interfaces with React, WebSockets, and Leaflet.js, rendering 50,000+ dynamic charging points at sub-300ms latency, enabling live location tracking, status (available/charging/fault) visualization.
- Implemented RESTful and GraphQL APIs using Node.js and Python FastAPI to serve frontend clients, handling complex queries and real-time state synchronization.
- Engineered CI/CD pipelines (GitHub Actions) with zero-downtime deployments, integrated Playwright E2E testing covering 80%+ user journeys, achieving 95% pass rates and reducing production bugs by 20%.
- Enforced OAuth2, OIDC, JWT authentication flows, improving access management efficiency by 40%.
- Drove Gen AI adoption across product teams, accelerating feature delivery by 50% via intelligent tooling (Copilot, Cursor, n8n).

Software Developer Bosch Rexroth AG

04/2020 – 06/2022
Ulm, Germany

- Delivered 6 full-stack applications using Angular, React, TypeScript, Node.js, and Python for 600+ global partners, integrating REST/SOAP APIs and real-time data feeds to improve product transparency and increase partner onboarding efficiency by 16% within 9 months.
- Automated partner onboarding through API-driven workflows, reducing onboarding time from 2 weeks to 3 days and increasing operational productivity by 23%.
- Engineered an Angular/.NET Core Product Selector with SAP integration, 3D asset visualization, and real-time product configuration, deployed to 100+ partners, reducing configuration time from 2 hours to 30 minutes per setup.
- Architected a real-time IoT video integration platform streams with <150 ms latency, adopted by 600+ global partners, enabling reliable low-latency hardware-software communication and distributed real-time collaboration.
- Led the migration to Azure-based microservices (20-service decomposition) and implemented CI/CD pipelines using Jenkins and Docker, reducing deployment time from 4 hours to 36 minutes, lowering infrastructure costs by 40%, and improving system availability to 99.8% uptime.
- Built a Vue.js component library aligned with a Figma design system, delivering 45+ reusable dashboard and visualization components adopted by 4 product teams, reducing code redundancy by 35% and increasing sprint velocity by 18%.

CONTACT DETAILS

- 🌐 www.shubhamnarkhede.com
- 📍 Stuttgart, 70376 Germany
- ☎ +49 171 510 8059
- ✉ shubhampnarkhede@gmail.com
- 🌐 [LinkedIn.com/spnarkhede](https://www.linkedin.com/company/spnarkhede)
- 🐙 [GitHub.com/spnarkhede](https://github.com/spnarkhede)

SKILLS

- **Frontend:** React, Vue.js, Angular, TypeScript, JavaScript (ES6+), Redux, Pinia, HTML5/CSS3, Tailwind CSS
- **AI & Security:** LLM, n8n, Claude, Gemini, Copilot, Cursor, SSO, OAuth, JWT, OpenID, WebAuthn
- **Backend & APIs:** Node.js, Python (FastAPI, Flask), RESTful APIs, GraphQL, Microservices, PostgreSQL, MongoDB, Redis, RabbitMQ, MySQL, WebSockets, Socket.io, gRPC, OCPP
- **Designer:** Figma, Sketch, Adobe XD, Balsamiq, Canva, Miro, Notion, Wireframing, Prototyping
- **Testing & Quality:** Playwright, Jest, Vitest, E2E testing, unit testing, TDD, code reviews, WCAG accessibility, Postman, Swagger
- **DevOps & CI/CD:** Azure DevOps, GitHub Actions, Jenkins, Docker, Kubernetes, AWS, Grafana, Git
- **Agile & Collaboration:** Scrum, Kanban, Jira, Confluence, Figma, design systems, Docusaurus, InnerSource, VitePress

LANGUAGES

- English - Fluent (C1)
- German - Intermediate (B1)
- Hindi - Native
- Marathi - Mother tongue

EXPERTISE

Full Stack Development
3D Web Visualization
Design & Architecture
Real-Time Data Streaming
Real-Time Communication & Protocols
Database & Data Management
Testing & Quality Assurance
Agile & Collaboration

CERTIFICATIONS

Introduction to Project Management
AWS Fundamentals Specialization
AWS: Building Serverless Application
Python for Data Science
Introduction to Data Science
For More: <https://tinyurl.com/1000cert>

PERSONAL PASSIONS

AI Tools Exploration
Books Reading & Reviews Writing
UX/UI Designing & Prototyping
Online Learning & Certifications
Knowledge Summarization
System Prompts & Guide Creation

LIFE BEYOND WORK

Team Events Organizing
Art (Sketching & Painting)
Fitness (Workout, Yoga)
Travelling & Exploring Cultures
Sports & Activities
Photography & Nature Exploration
Musical Instrument Playing

Software Developer - Associate
Amazon
04/2019 – 03/2020
Leipzig, Germany

- Delivered scalable internal tools (TypeScript, Node.js, Java, Spring Boot), improving operational efficiency by 30% across fulfillment and logistics operations.
- Designed RESTful APIs handling high-throughput workflows, applied test-driven development (Jest, unit testing) and code reviews (50+ reviews) to reduce production incidents by 20%.
- Accelerated cross-functional delivery by 25% through agile sprint planning, collaborating with UX, product, and engineering teams on feature prioritization and rapid prototyping.

EDUCATION

M.Sc. Automotive Software Engineering
Chemnitz University of Technology
Oct 2018 – Aug 2020
Chemnitz, Germany

B.E. Computer Science & Engineering
Rashtrasant Tukadoji Maharaj Nagpur University
Jun 2014 – May 2018
Nagpur, India

INDUSTRIAL PROJECTS

Connected Charging Cable Platform
Angular | TypeScript | Vite | REST APIs | OCPP | Docker | AWS
Real-time operator interface processing 10K+ OCPP messages/sec at sub-200ms latency across 8 European markets.

Charge Point Management System (CPMS)
React | Vue.js | TypeScript | PHP (Symfony) | MySQL | Redis | RabbitMQ | Docker | Kubernetes | AWS
Micro-frontend architecture dashboards and Symfony backend for 450,000+ charge points across 30 countries with 99.9% uptime.

Real-Time Geospatial Dashboard
React | TypeScript | Leaflet.js | REST APIs | WebSockets
Interactive map rendering 50K+ data points sub-300ms render time, 60fps pan/zoom with dynamic filtering.

High-Volume Data Visualization Platform
React | TypeScript | D3.js | WebSockets | Canvas API
10K+ data points/second charting at 60fps using canvas rendering, deployed across 8 enterprise products.

Application Product Selector (APS)
Vue.js | Pinia | TypeScript | Vite | PHP (Symfony) | MySQL | 3D MAPs | SAP
SAP-connected product catalog with 100+ global partners, 35% faster configuration, 200+ manual requests eliminated monthly.

Smart Project Management v3.0
Angular | TypeScript | PHP | MySQL | Azure | Kubernetes | Docker
Multi-tenant project management system for 1,000+ users, reducing project cycle time by 22%.

Support Portal 2.0
React | Vue.js | TypeScript | OAuth | REST APIs | Figma
Internal support platform 40% faster ticket resolution, 60% of L1 tasks automated for 300+ users.